

Anti-Bullying Reporting Platform: Complete Project Blueprint

Project: Anonymous Bullying Reporting & Case Management System
Client Delivery: Premium, production-grade web application
Status: Final Plan, Awaiting Approval

1. Executive Summary

A secure, anonymous bullying-reporting platform that empowers individuals to report incidents without fear of retaliation. The system provides a robust administrative dashboard for case management, analytics, and resolution tracking, all wrapped in a world-class, premium visual experience.

Key Differentiators:

- Zero-friction anonymous reporting (no account required)
- Unique Report ID with real-time status tracking
- Two-way anonymous communication between admin and reporter
- Enterprise-grade security (RLS, RBAC, rate-limiting, CSP)
- Fully responsive across all devices (mobile, tablet, desktop, ultrawide)
- Intelligent data lifecycle management with automated storage optimization
- Premium dark-mode glassmorphism design with micro-animations

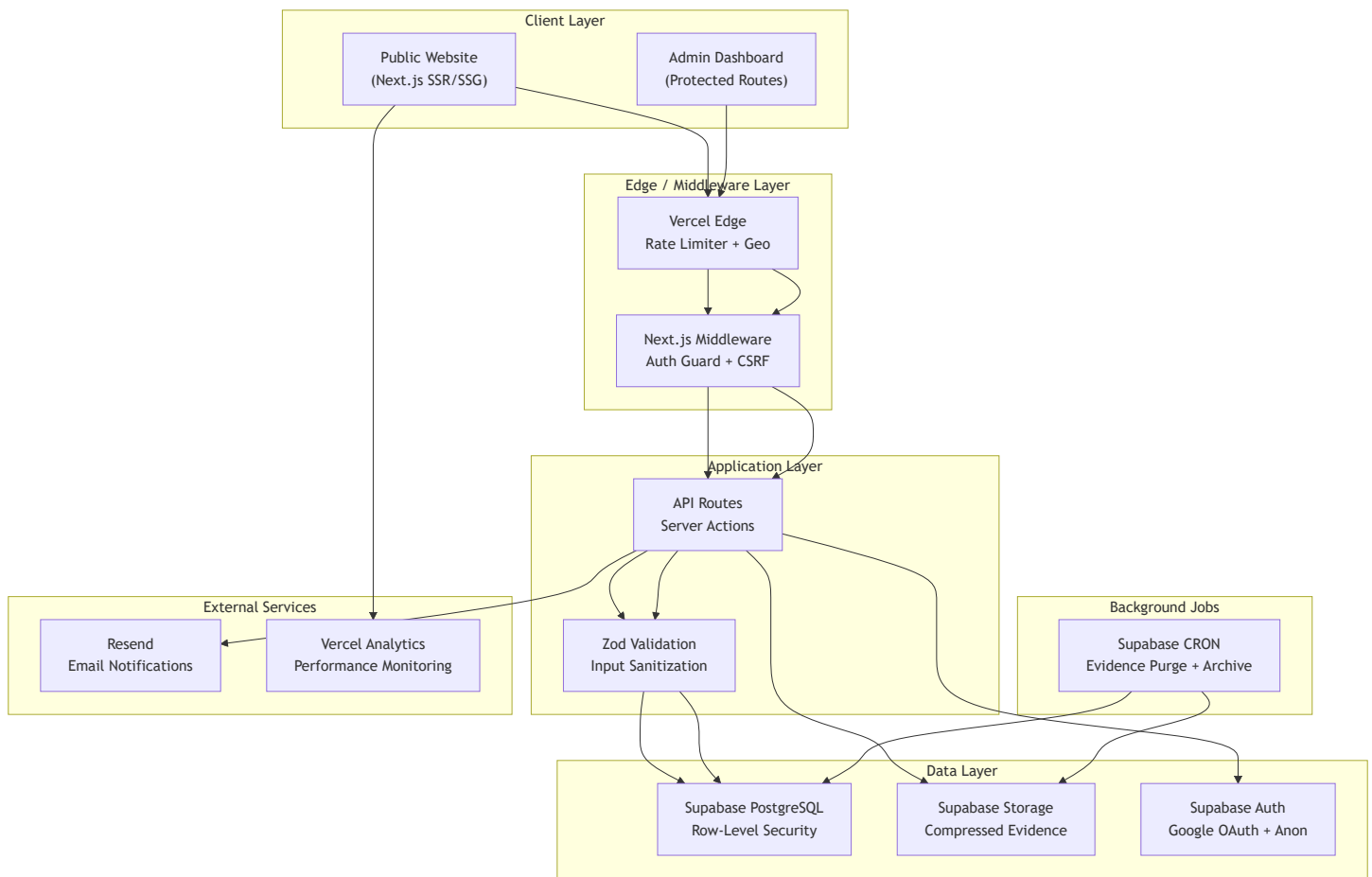
2. Technology Architecture

2.1 Tech Stack

Layer	Technology	Purpose
Frontend	Next.js 14+ (App Router)	Server-side rendering, SEO, routing

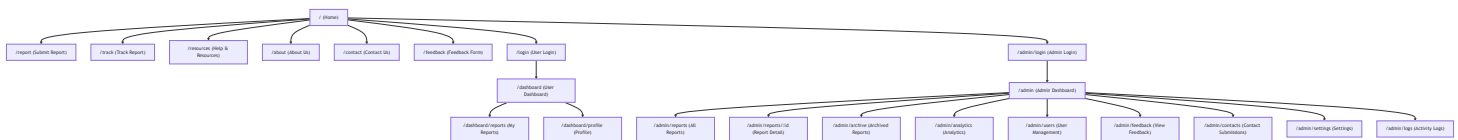
Layer	Technology	Purpose
Styling	Vanilla CSS + CSS Variables	Design system, animations, responsiveness
Backend	Next.js API Routes + Server Actions	Business logic, server-side validation
Database	Supabase (PostgreSQL)	Data storage, Row-Level Security, real-time subscriptions
Authentication	Supabase Auth (Google OAuth + Anonymous Sign-In)	User authentication
File Storage	Supabase Storage	Evidence uploads (images, screenshots)
Deployment	Vercel	Hosting, edge functions, CDN, SSL
Email	Supabase Edge Functions + Resend	Admin notifications, status update alerts
Validation	Zod	Schema-based input validation
Animations	Framer Motion	Premium micro-interactions
Icons	Lucide React	Consistent, modern iconography
Fonts	Google Fonts (Inter / Plus Jakarta Sans)	Premium typography
Image Processing	browser-image-compression	Client-side resize and WebP conversion before upload

2.2 High-Level Architecture



3. Site Structure & Navigation

3.1 Sitemap



3.2 Page-by-Page Breakdown

#	Page	Access Level	Description
1	Home /	Public	Hero section, how-it-works, statistics counter, CTA to report

#	Page	Access Level	Description
2	Submit Report /report	Public	Multi-step report form with anonymous or authenticated submission
3	Track Report /track	Public	Enter Report ID to view status timeline
4	Resources /resources	Public	Helpline numbers, anti-bullying guides, educational content
5	About Us /about	Public	Mission, team, vision (name/logo TBA)
6	Contact Us /contact	Public	Contact form with admin phone numbers, social links
7	Feedback /feedback	Public	Anonymous feedback form about the platform
8	Login /login	Public	Google OAuth login for users who want to track reports
9	User Dashboard /dashboard	Authenticated User	View submitted reports, status updates, profile management
10	Admin Login /admin/login	Admin Only	Admin-specific authentication
11	Admin Dashboard /admin	Admin Only	Overview cards, recent reports, quick actions
12	Report Management /admin/reports	Admin Only	Filterable, searchable report table with bulk actions
13	Report Detail /admin/reports/:id	Admin Only	Full report view, status updates, notes, communication log
14	Archived Reports /admin/archive	Admin Only	Resolved/closed cases (text-only, evidence purged)
15	Analytics /admin/analytics	Admin Only	Charts, trends, category breakdown, resolution metrics
16	User Management /admin/users	Admin Only	View registered users, manage roles

#	Page	Access Level	Description
17	Feedback Viewer /admin/feedback	Admin Only	All submitted feedback in sortable table
18	Contact Submissions /admin/contacts	Admin Only	All contact form submissions
19	Activity Logs /admin/logs	Admin Only	Timestamped audit trail of all admin actions
20	Settings /admin/settings	Admin Only	Platform configuration, notification preferences

4. User Features (Complete Breakdown)

4.1 Anonymous Reporting (No Account Required)

Feature	Details
Multi-Step Report Form	Guided wizard: Incident Type > Details > Evidence > Security > Review > Submit
Report Types	Verbal, Physical, Cyberbullying, Social/Relational, Sexual, Other
Severity Levels	Low, Medium, High, Critical (user selects)
Incident Details	Date/time of incident, location, description (mandatory, min 50 chars), involved parties (optional)
Security Passphrase	Min 6 characters, required for tracking. Hashed via bcrypt before storage; plaintext never persisted
Evidence Upload	Up to 3 files (images, screenshots), max 5MB each, auto-compressed to WebP before upload
Witness Information	Optional: names or details of witnesses
Unique Report ID	System generates a human-readable tracking code (e.g., RPT-2026-A7X3K9) with cryptographic uniqueness

Feature	Details
Report Receipt	Success screen with Report ID, option to copy/download, QR code for quick access
No IP Logging	Reporter's IP address is never stored in the reports database
Draft Saving	Auto-save to localStorage so users don't lose progress on accidental close. Drafts expire after 7 days
Idempotency Protection	Each submission includes a unique client-generated key to prevent accidental double-submits

4.2 Authenticated User Features (Google Login)

Feature	Details
Google OAuth Login	One-click login via Google, no passwords to manage
Personal Dashboard	View all submitted reports with status, timestamps, and filters
Report History	Chronological list of all reports with search and filter
Status Notifications	Real-time badge/indicator when admin updates a report status
Two-Way Messaging	Anonymous chat thread with admin on each report (identity still protected if anonymous option was chosen)
Edit/Withdraw Report	User can request withdrawal of a report before admin marks it as "Under Review"
Profile Management	View/edit display name, notification preferences
Data Export	Download all personal reports as a JSON/CSV file
Account Deletion	User can request account deletion; orphans anonymous reports, deletes profile data
Logout	Secure session termination

4.3 Report Tracking (Public, No Login Needed)

Feature	Details
Track by Report ID	Enter the Report ID on the public tracking page
Visual Status Timeline	Animated vertical timeline showing all status transitions
Status States	Submitted > Received > Under Review > Investigation > Action Taken > Resolved / Closed / Dismissed
Admin Notes (Public)	Admin can attach public-facing notes visible to the reporter
Estimated Resolution	Optional field admins can set to communicate expected timelines
Report Verification	Requires Report ID + a security passphrase (set during submission) for added privacy
Archived Report View	Resolved reports show a condensed text-only summary (evidence auto-purged after 30 days)
Archived Messaging	Messaging is disabled on archived reports. Reporter sees "This case is closed. No further messages can be sent."

4.4 Contact Us

Feature	Details
Contact Form	Name, Email, Subject, Message with validation
Admin Contact Info	Display admin names and phone numbers (as provided)
Social Media Links	Placeholders for Instagram, Twitter/X, Facebook, LinkedIn (TBA)
Map Embed	Optional: Google Maps embed for physical address
Auto-Response	Success toast notification on form submission
Rate Limiting	Max 3 contact submissions per IP per hour to prevent spam

4.5 Feedback Form

Feature	Details
Anonymous Feedback	No login required
Rating System	5-star rating for overall platform experience
Category Tags	UI/UX, Report Process, Speed, Communication, Other
Free-Text Field	Detailed comments (max 2000 characters)
Submission Confirmation	Animated success state
Rate Limiting	Max 5 feedback submissions per IP per day to prevent spam

4.6 Resources Page

Feature	Details
Helpline Directory	National/local anti-bullying helpline numbers
Educational Content	What is bullying, types, how to respond, bystander tips
FAQ Accordion	Common questions with smooth expand/collapse
External Links	Links to verified organizations and support groups

5. Admin Panel Features (Complete Breakdown)

5.1 Admin Authentication & Access Control

Feature	Details
Admin Login	Google OAuth restricted to whitelisted email addresses
Role-Based Access	Super Admin (full access) and Admin (case management only)
Session Management	Auto-logout after 30 minutes of inactivity

Feature	Details
Admin Whitelist	Only pre-approved emails can access admin panel
Activity Logging	Every admin action is logged with timestamp, user, and action type
Session Revalidation	Middleware checks admin status on every request; if removed mid-session, immediate logout

Initial Admin List:

Name	Phone	Role
Poorvi Aggarwal	88827 03319	Super Admin
Shreya Aggarwal	8882962300	Admin
Utkarsh Lohan	85958 69833	Admin
Rituraj Sharma	8882904396	Admin

5.2 Dashboard Overview

Feature	Details
Stats Cards	Total Reports, Pending, In Progress, Resolved, Dismissed (animated counters)
Recent Reports	Last 10 reports with quick-action buttons
Trend Line Chart	Reports over time (last 7/30/90 days)
Category Pie Chart	Distribution of report types
Severity Heat Map	Visual indicator of incoming severity levels
Quick Actions	Jump to: New Reports, Flagged, Overdue items
Storage Usage Meter	Visual indicator of current storage consumption (evidence files)
Empty State	If no reports exist yet, show an illustrated "all clear" message instead of blank tables

5.3 Report Management

Feature	Details
Report Table	Sortable columns: ID, Date, Type, Severity, Status, Last Updated
Pagination	Server-side cursor-based pagination, 20 items per page, with page navigation
Advanced Filters	By status, severity, type, date range, keyword search
Bulk Actions	Select multiple > Change status, Export, Archive
Report Detail View	Full incident details, evidence gallery, timeline, admin notes
Status Update	Dropdown to change status with mandatory note explaining the change
Priority Flagging	Flag reports as urgent with visual indicator
Internal Notes	Admin-only notes not visible to the reporter
Public Notes	Notes visible to the reporter on the tracking page
Evidence Viewer	In-panel image viewer with zoom, download option. Shows "Evidence purged" label for archived reports
Delete Report	Soft-delete with confirmation modal (moves to archive, recoverable for 30 days)
Permanent Delete	Super Admin only, irreversible deletion with double confirmation
Export Report	Export individual report as PDF
Print Report	Formatted print view for physical records
Merge Reports	Link duplicate/related reports under a parent report

5.4 Two-Way Communication

Feature	Details
Message Thread	Per-report chat-like interface between admin and reporter
Admin Identity Shown	Admin name visible to reporter in messages
Reporter Stays Anonymous	Reporter identity never exposed even in authenticated mode (if anonymous option was selected)
Message Notifications	Badge indicator for unread messages
Message History	Full scrollable history with timestamps
Message Length Limit	Max 2000 characters per message to prevent abuse
Message Retention	Messages for resolved cases are archived as text, not purged

5.5 Analytics Dashboard

Feature	Details
Reports Over Time	Line chart with daily/weekly/monthly granularity
Category Breakdown	Doughnut chart of bullying types
Severity Distribution	Bar chart of severity levels
Resolution Rate	Percentage of reports resolved vs. total
Average Resolution Time	How long from submission to resolution
Peak Reporting Times	Heat map of day-of-week x hour-of-day
Status Funnel	Visual funnel from Submitted to Resolved
Date Range Picker	Filter all analytics by custom date range
Export Data	CSV/PDF export of analytics data
Empty State	Friendly "Not enough data yet" placeholder when data is insufficient for charts

5.6 User Management

Feature	Details
Registered Users List	Email, display name, join date, report count (paginated)
User Details	View user's report history (without revealing anonymous reports)
Block User	Prevent spam accounts from submitting
Admin Role Management	Super Admin can promote/demote admins
Last Active	Timestamp of last login for each user

5.7 Feedback & Contact Management

Feature	Details
Feedback Table	Rating, category, message, date, sortable and filterable (paginated)
Contact Submissions	Name, email, subject, message, with read/unread status (paginated)
Reply to Contact	Open email client with pre-filled reply
Mark as Read/Resolved	Status tracking for contact submissions
Bulk Delete	Remove old feedback/contact entries in bulk

5.8 Activity Logs & Audit Trail

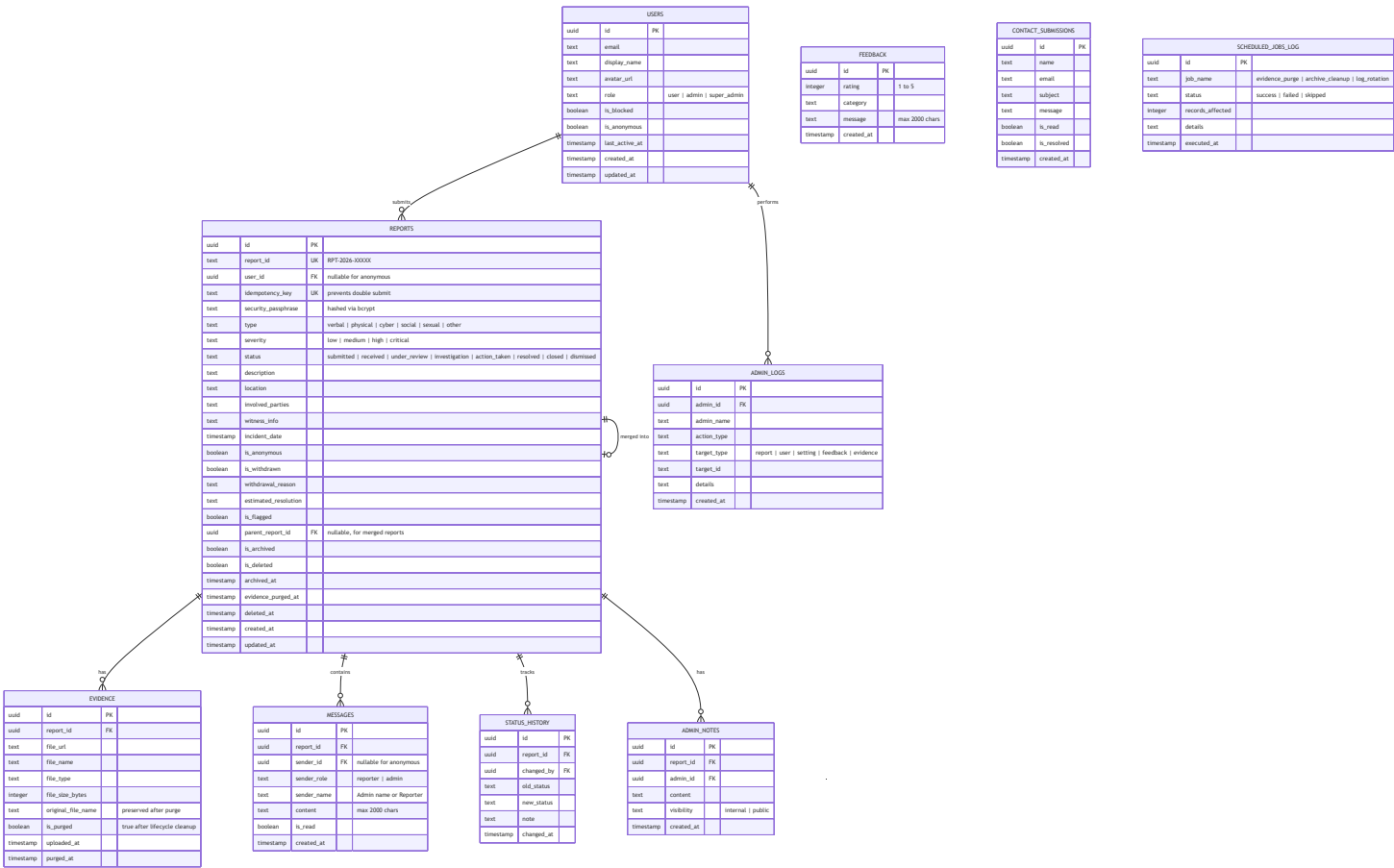
Feature	Details
Action Log	Every admin action logged: status changes, deletions, role changes, evidence purges
Log Fields	Timestamp, Admin Name, Action Type, Target (Report ID/User), Details
Filterable	By admin, action type, date range
Non-Deletable	Logs cannot be deleted or modified, integrity guaranteed
Retention	Logs retained for 1 year, then auto-archived to compressed JSON export
Pagination	Server-side pagination with infinite scroll

5.9 Settings

Feature	Details
Notification Preferences	Toggle email notifications for new reports, status changes
Platform Config	Site name (TBA), logo upload, social media links
Report Categories	Add/edit/remove bullying categories
Status Workflow	Customize available statuses and transitions
Evidence Retention	Configure retention period for evidence files after resolution (default: 30 days)
Maintenance Mode	Toggle site into maintenance with custom message
Storage Dashboard	View current database and file storage consumption

6. Database Schema

6.1 Entity-Relationship Diagram



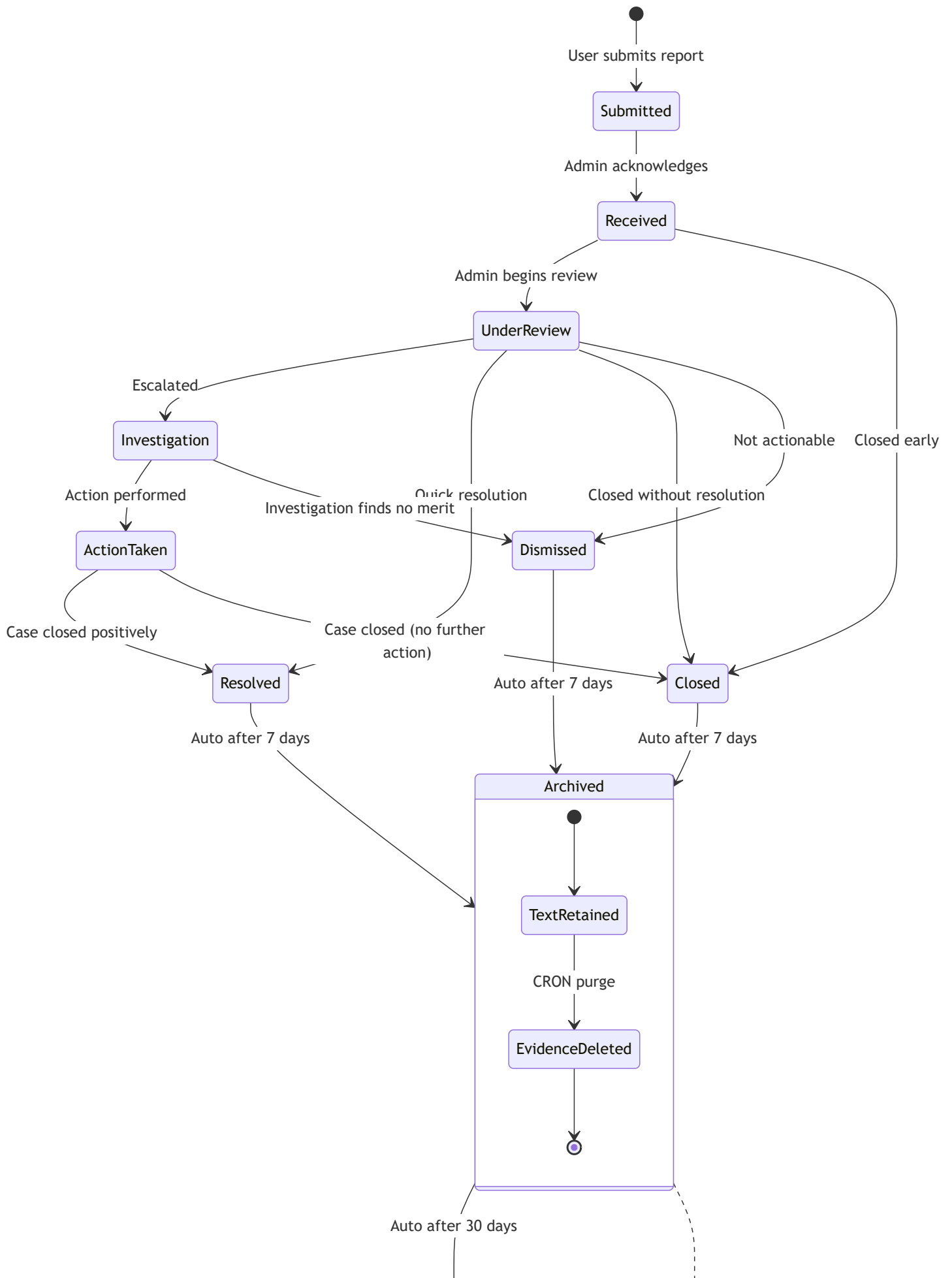
6.2 Key Indexes

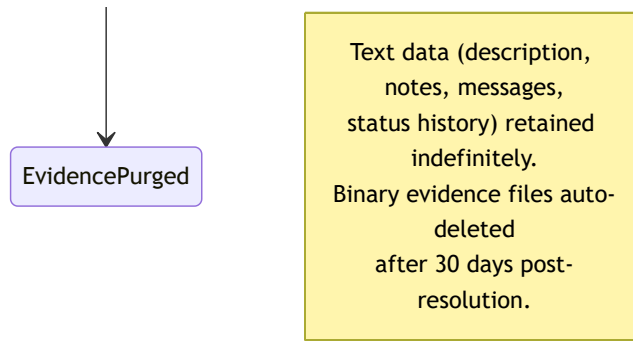
Table	Column(s)	Purpose
reports	report_id (UNIQUE)	Fast lookup by tracking ID
reports	idempotency_key (UNIQUE)	Prevent duplicate submissions
reports	user_id	RLS policy performance
reports	status, is_deleted, is_archived	Composite filter for active reports
reports	created_at	Sort by date
reports	parent_report_id	Merged report lookup
evidence	report_id, is_purged	Evidence retrieval with purge awareness

Table	Column(s)	Purpose
messages	report_id, created_at	Message thread ordering
status_history	report_id, changed_at	Timeline rendering
admin_logs	created_at	Log browsing
admin_logs	action_type	Filter by action type

7. Data Lifecycle & Storage Optimization

7.1 Report Lifecycle State Machine

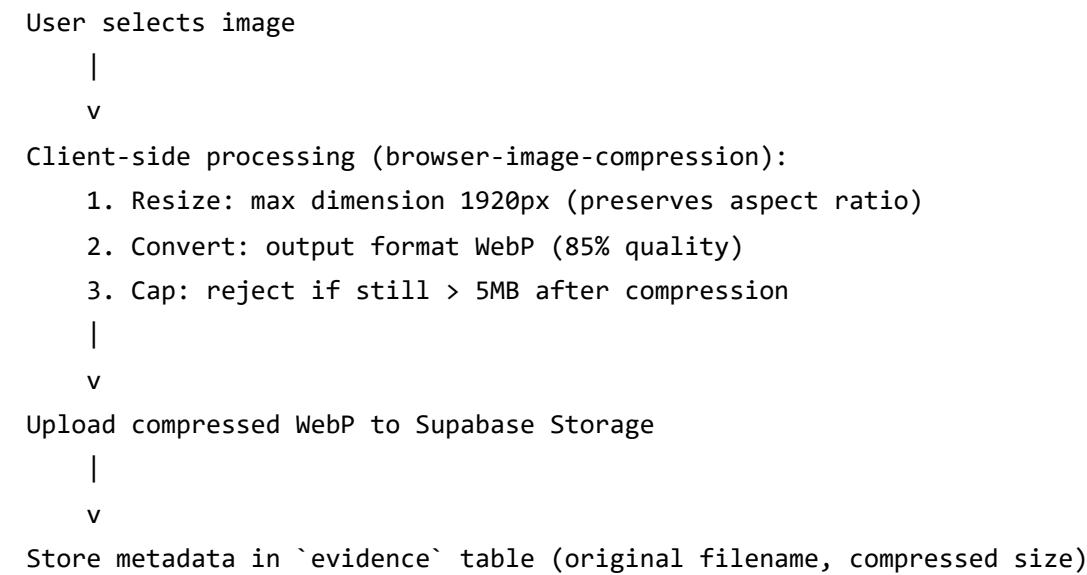




7.2 Evidence Lifecycle

Phase	Timeframe	What Happens
Active	Report status is NOT resolved/closed/dismissed	Evidence files are stored and fully accessible to admins
Grace Period	0 to 30 days after resolution	Evidence still accessible. Admin can manually download/export before purge
Purge	30 days after resolution (CRON job)	Binary files deleted from Supabase Storage. evidence row kept with <code>is_purged = true</code> , preserving metadata (filename, type, size, upload date)
Indefinite	Forever	Text data (report description, notes, messages, status history, admin logs) retained permanently

7.3 Image Compression Pipeline



Storage savings estimate: Average phone photo (4-8MB JPEG) compresses to 200-500KB WebP. This is a 10-15x reduction.

7.4 Automated Cleanup Jobs (Supabase CRON via pg_cron)

Job	Schedule	Action
Evidence Purge	Daily at 02:00 UTC	Find reports where status IN (resolved, closed, dismissed) AND archived_at < NOW() - 30 days AND evidence_purged_at IS NULL . Delete storage files, set evidence.is_purged = true , set reports.evidence_purged_at = NOW() . Log to scheduled_jobs_log
Soft-Delete Cleanup	Weekly (Sunday 03:00 UTC)	Permanently delete reports where is_deleted = true AND deleted_at < NOW() - 30 days . Remove all associated evidence, messages, notes, and status history. Log results
Auto-Archive	Daily at 01:00 UTC	Move reports with terminal status (resolved/closed/dismissed) older than 7 days to is_archived = true
Stale Draft Notification	Weekly	Identify reports in submitted status for > 14 days with no admin action; flag them in dashboard
Log Rotation	Monthly (1st, 04:00 UTC)	Archive admin logs older than 12 months to compressed JSON export, then delete originals
Anonymous Session Cleanup	Weekly	Remove orphaned anonymous auth sessions older than 90 days with no associated reports

7.5 Storage Budget Awareness

Resource	Budget-Conscious Approach
Database rows	Cursor-based pagination on all queries (no OFFSET). Composite indexes on frequently filtered columns. EXPLAIN ANALYZE on critical queries
File storage	Auto-purge lifecycle (Section 7.2). Client-side compression before upload. Max 3 files per report, 5MB cap each (post-compression)

Resource	Budget-Conscious Approach
Bandwidth	Next.js Image Optimization serves responsive sizes via <Image> component. Static pages use ISR with 1-hour revalidation. CDN caching via Vercel edge
Realtime connections	Subscribe only on pages that need it (admin detail view, user dashboard). Unsubscribe on unmount. Single channel per report, not per-table
Auth sessions	Anonymous sessions auto-cleaned after 90 days. Session tokens stored in HttpOnly cookies (no localStorage bloat)
Email volume	Batch notifications: if 5+ reports arrive within 10 minutes, send one digest email instead of 5 separate ones. Weekly analytics digest is a single email, not per-metric

7.6 Data Retention Policy

Data Type	Retention Period	After Expiry
Report text data	Indefinite	Kept permanently for historical records and analytics
Evidence files	30 days after resolution	Binary files deleted; metadata row preserved
Messages	Indefinite	Text retained as part of case record
Status history	Indefinite	Preserved for audit and analytics
Admin logs	12 months active, then archived	Exported as compressed JSON, then originals deleted
Feedback	Indefinite	Lightweight text data, minimal storage cost
Contact submissions	6 months after marked resolved	Auto-deleted by CRON
Soft-deleted reports	30 days	Permanently deleted with all associated data
Anonymous auth sessions	90 days (no reports)	Cleaned by CRON

Data Type	Retention Period	After Expiry
localStorage drafts	7 days	Client-side TTL, auto-cleared

8. Resource-Efficient Architecture

This section documents the engineering principles used to ensure the platform operates with maximum efficiency at any scale, from initial launch to high-traffic growth, without requiring code-level changes when scaling resources.

8.1 Stateless, Horizontally Scalable Design

- All server-side logic is stateless (no in-memory sessions, no server-local caches)
- Authentication state is managed via signed JWTs and HTTP-only cookies
- Any Vercel serverless function instance can handle any request independently
- Database connection pooling via Supavisor (Supabase's built-in connection pooler) prevents connection exhaustion under concurrent load

8.2 Edge-First Computation

- Rate limiting and auth guard checks run in Vercel Edge Middleware (V8 isolates, sub-millisecond cold start)
- Static and ISR pages are served from the CDN edge node closest to the user
- API routes use Vercel Serverless Functions with automatic regional deployment

8.3 Lazy Evaluation & Minimal Data Transfer

- All list views use server-side cursor-based pagination (never `SELECT * without LIMIT`)
- Admin dashboard stats computed via PostgreSQL aggregate queries (COUNT, AVG) rather than fetching all rows to the client
- Analytics charts use database-level GROUP BY with date truncation, not client-side data processing
- Evidence images are lazy-loaded and served as optimized WebP thumbnails
- Supabase Realtime subscriptions are scoped to individual report channels, not table-wide listeners

8.4 Intelligent Caching Strategy

Content Type	Cache Strategy	TTL
Home / About / Resources	Static Generation (SSG)	Revalidate every 1 hour (ISR)
Report Form	Client-side render (CSR)	No cache (dynamic)
Tracking Page	Server-side render (SSR)	No cache (real-time data)
Admin Dashboard	SSR with stale-while-revalidate	30 seconds
Analytics Charts	SSR with ISR	5 minutes
Static assets (CSS, JS, fonts)	Immutable CDN cache	1 year (hash-busted filenames)

8.5 Database Query Optimization

- All RLS policy auth functions wrapped in `(SELECT auth.uid())` subquery to prevent per-row re-evaluation
- Composite indexes on `(status, is_deleted, is_archived)` for the most common admin query pattern
- `EXPLAIN ANALYZE` run on all critical queries during development to verify index usage
- Partial indexes where applicable (e.g., index only non-deleted, non-archived reports)
- Connection pooling mode: `transaction` (returns connection to pool after each query)

8.6 Email Efficiency

- Digest batching: multiple events within a 10-minute window consolidated into a single notification
- Weekly analytics sent as one summary email, not per-metric
- Email templates are static HTML (no server-side rendering per send)
- Bounce handling: failed emails logged but not retried endlessly (max 2 retries)

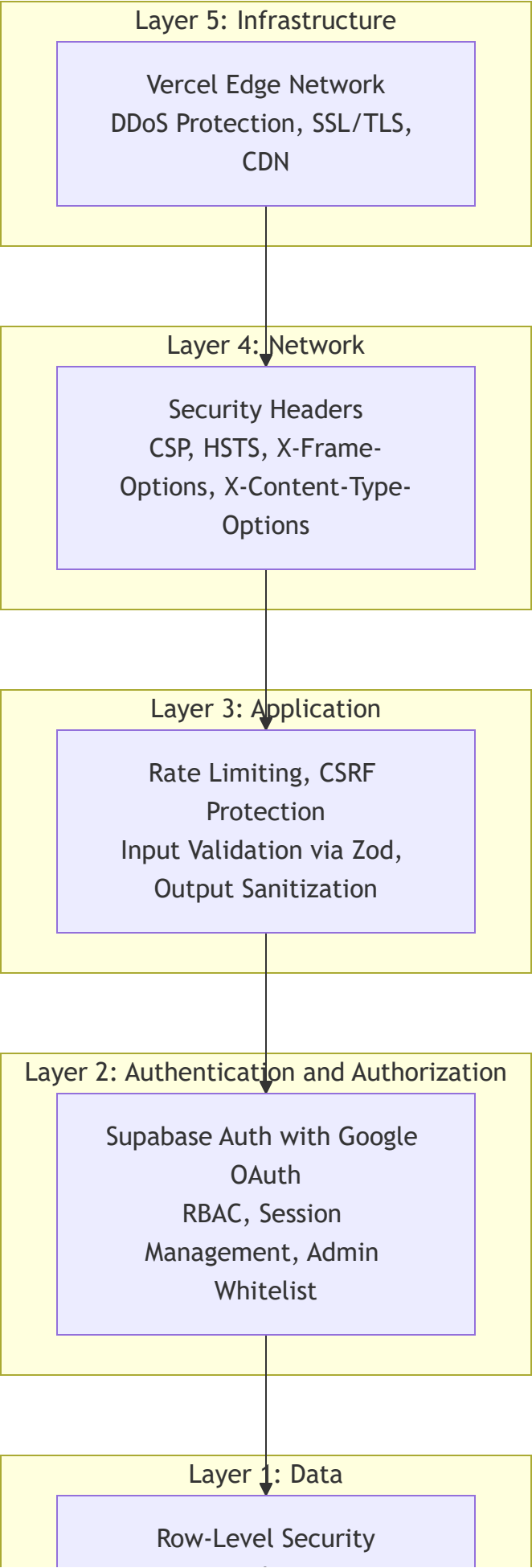
8.7 Graceful Degradation

Failure Scenario	Behavior
Email service down	Reports still accepted; admin notifications queued; toast shown "Notification delivery delayed"

Failure Scenario	Behavior
Supabase Realtime unavailable	Dashboard falls back to polling (30-second interval)
Google OAuth temporarily down	Login page shows "Authentication service temporarily unavailable" with retry button
Storage quota reached	New uploads blocked with clear message "Evidence upload temporarily unavailable"; admin alerted via dashboard banner
Database connection pool exhausted	Requests queue with 5-second timeout; 503 returned with "Please try again in a moment"

9. Security Architecture

9.1 Security Pyramid



Encrypted Storage,
Parameterised Queries
Soft-Delete, Immutable
Audit Logs

9.2 Attack Mitigation Matrix

Attack Vector	Mitigation	Implementation
SQL Injection	Parameterised queries	Supabase client uses parameterised queries by default; no raw SQL on client
XSS	CSP headers + React auto-escaping + DOMPurify	<code>next.config.js</code> CSP policy; sanitize any user-rendered HTML
CSRF	Server Actions (POST-only) + SameSite cookies	Next.js Server Actions; <code>SameSite=Strict</code> on auth cookies
Broken Authentication	Google OAuth only (no password storage)	Supabase Auth with Google provider; no custom auth logic
Broken Access Control	RLS + Middleware guards + RBAC	RLS policies on every table; middleware checks role before admin routes
Rate Limiting	Edge middleware throttling	IP-based rate limit: 30 req/min general, 5/min for submissions, 5/hour for tracking attempts
File Upload Abuse	Type validation + size limits + client-side compression	Accept only <code>image/png</code> , <code>image/jpeg</code> , <code>image/webp</code> ; max 5MB post-compression; Supabase Storage policies
Data Exposure	Minimal data in API responses + RLS	Server-side filtering; never return admin-only fields to public endpoints
Clickjacking	X-Frame-Options: DENY	Security headers in <code>next.config.js</code>
Brute Force (Report ID)	Security passphrase + rate limit on	Report tracking requires ID + passphrase; 5 attempts per IP per hour, then 1-hour lockout

Attack Vector	Mitigation	Implementation
	tracking	
Spam Reports	Honeypot fields + rate limiting + CAPTCHA fallback	Hidden form field; submission cooldown; hCaptcha triggered if >3 submissions from same IP in 1 hour
Session Hijacking	HttpOnly + Secure + SameSite cookies	Supabase session management defaults
Privilege Escalation	Server-side role checks on every admin action	Middleware + API route level RBAC validation; role checked from DB, not from JWT alone
Data Tampering	Immutable audit logs	<code>admin_logs</code> table: insert-only RLS policy, no UPDATE/DELETE allowed
Report ID Enumeration	Cryptographically random IDs	Report IDs use <code>crypto.randomBytes(6).toString('hex')</code> prefixed with <code>RPT-YYYY-</code> , yielding 281 trillion combinations
Storage Abuse	Per-report file limits + compression + lifecycle purge	Max 3 files/report, 5MB each, auto-purge after 30 days post-resolution
Replay Attack	Idempotency keys on submissions	Each form submission includes a UUID idempotency key; server rejects duplicates
CRON Endpoint Abuse	Shared secret authentication	All <code>/api/cron/*</code> endpoints require <code>Authorization: Bearer \$CRON_SECRET</code> header; reject with 401 if missing/invalid
Contact/Feedback Spam	Per-IP rate limits + honeypot	Contact: 3/hour per IP. Feedback: 5/day per IP. Hidden honeypot field on both forms

9.3 Row-Level Security Policies

reports table:

Anonymous users > INSERT only (via Supabase anonymous sign-in)

Authenticated users > INSERT + SELECT own reports only (WHERE user_id = auth.uid())

Admins > Full CRUD (except permanent delete = Super Admin only)

messages table:

Reporter > INSERT + SELECT on their own report thread

Admins > INSERT + SELECT on all report threads

admin_logs table:

Admins > INSERT only (no UPDATE, no DELETE)

Super Admin > SELECT all logs

evidence table:

Reporter > INSERT on their own report (max 3 files enforced at API level)

Admins > SELECT all evidence

admin_notes table:

Admins > INSERT + SELECT + UPDATE own notes

Super Admin > SELECT all notes

feedback / contact_submissions:

Public > INSERT only

Admins > SELECT + UPDATE (mark as read)

Super Admin > DELETE

scheduled_jobs_log:

Service role only > INSERT (via pg_cron)

Super Admin > SELECT (view job history)

10. Design System & Visual Identity

10.1 Design Philosophy

Principle	Implementation
Dark-First	Deep charcoal base (#0A0A0F), not pure black; reduces eye strain and feels safe
Glassmorphism	Frosted-glass cards with backdrop-filter: blur(20px) and subtle borders
Bold Gradients	Primary accent gradient: deep violet to electric blue to soft teal
Micro-Animations	Framer Motion for page transitions, hover states, and loading skeletons
Typography-Driven	Inter (body) + Plus Jakarta Sans (headings), clean, modern, highly legible
Accessibility	WCAG 2.1 AA compliant: contrast ratios, keyboard navigation, screen reader support, focus indicators
Bento Grid Layout	Modular card-based layouts for dashboard and resource pages

10.2 Colour Palette

Token	Hex	Usage
--bg-primary	#0A0A0F	Page background
--bg-secondary	#12121A	Card backgrounds
--bg-elevated	#1A1A2E	Elevated surfaces, modals
--bg-glass	rgba(255, 255, 255, 0.04)	Glassmorphism cards
--border-glass	rgba(255, 255, 255, 0.08)	Glass card borders

Token	Hex	Usage
--text-primary	#F0F0F5	Main text
--text-secondary	#8888A0	Muted text, labels
--accent-violet	#7C3AED	Primary actions
--accent-blue	#3B82F6	Secondary actions, links
--accent-teal	#14B8A6	Success states, confirmations
--accent-amber	#F59E0B	Warnings, in-progress states
--accent-red	#EF4444	Errors, critical severity, destructive actions
--gradient-primary	linear-gradient(135deg, #7C3AED, #3B82F6, #14B8A6)	CTAs, hero elements

10.3 Component Library

Component	Style
Buttons	Gradient fill (primary), ghost (secondary), outlined (tertiary), all with hover glow effect
Cards	Glassmorphism with blur, subtle border, hover lift animation
Inputs	Dark backgrounds with glowing border on focus (violet tint)
Modals	Centred overlay with backdrop blur and scale-in animation
Tables	Striped rows with hover highlight, sticky headers, responsive card view on mobile
Badges	Pill-shaped status badges with colour-coded backgrounds

Component	Style
Toasts	Slide-in from top-right with auto-dismiss and progress bar
Skeleton Loaders	Shimmering placeholder blocks during data fetches
Timeline	Vertical line with animated nodes for status tracking
Charts	Dark-themed with gradient fills and smooth curves
Sidebar (Admin)	Collapsible with icon-only mode, smooth transitions
Navbar	Transparent with blur background, logo left, actions right
Footer	Social links (TBA), quick links, legal text
Empty States	Custom illustrated messages for zero-data views
Error States	Friendly error pages with retry actions and contextual guidance
Pagination	Minimal design with prev/next + page numbers, keyboard navigable

10.4 Animations & Transitions

Element	Animation
Page Transitions	Fade + subtle slide-up (200ms)
Card Hover	Translate Y -4px + shadow expansion
Button Hover	Gradient shift + subtle glow
Form Steps	Slide-in from right
Status Timeline	Staggered node reveal on scroll
Dashboard Counters	Counting-up number animation on load
Modal Open	Scale from 0.95 to 1.0 + backdrop fade
Toast	Slide in from right + fade out
Loading States	Shimmer skeletons
Hero Section	Floating gradient orbs with slow drift animation

Element	Animation
Evidence Purge Indicator	Subtle strikethrough animation on purged file metadata

11. Responsive Design Strategy

11.1 Breakpoints

Breakpoint	Width	Layout
Mobile S	320px	Single column, stacked elements, bottom navigation
Mobile L	480px	Single column, slightly larger touch targets
Tablet	768px	Two-column grids, sidebar becomes hamburger menu
Desktop	1024px	Full layout, sidebar visible
Desktop L	1280px	Wider content area, more dashboard cards per row
Ultrawide	1536px+	Max-width container centred, extra breathing room

11.2 Mobile-Specific Adaptations

- Bottom sheet modals instead of centred modals
- Swipe gestures for report navigation
- Sticky submit button on report form
- Collapsible sections with touch-friendly toggles
- Responsive data tables become card view on mobile
- Touch-optimized slider for severity selection
- Reduced animation intensity on `prefers-reduced-motion` media query

11.3 Browser Compatibility

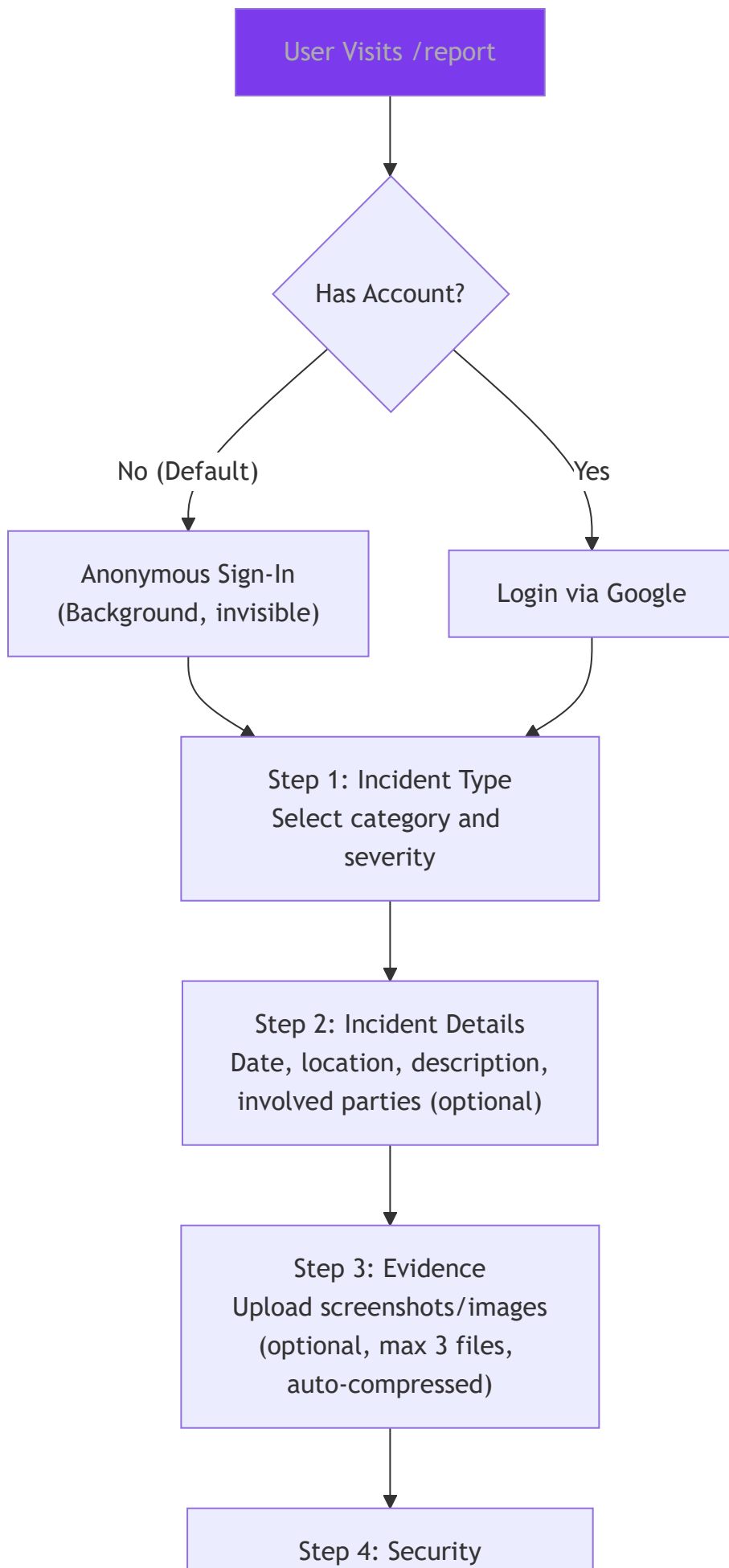
Browser	Minimum Version
Chrome	90+
Firefox	90+

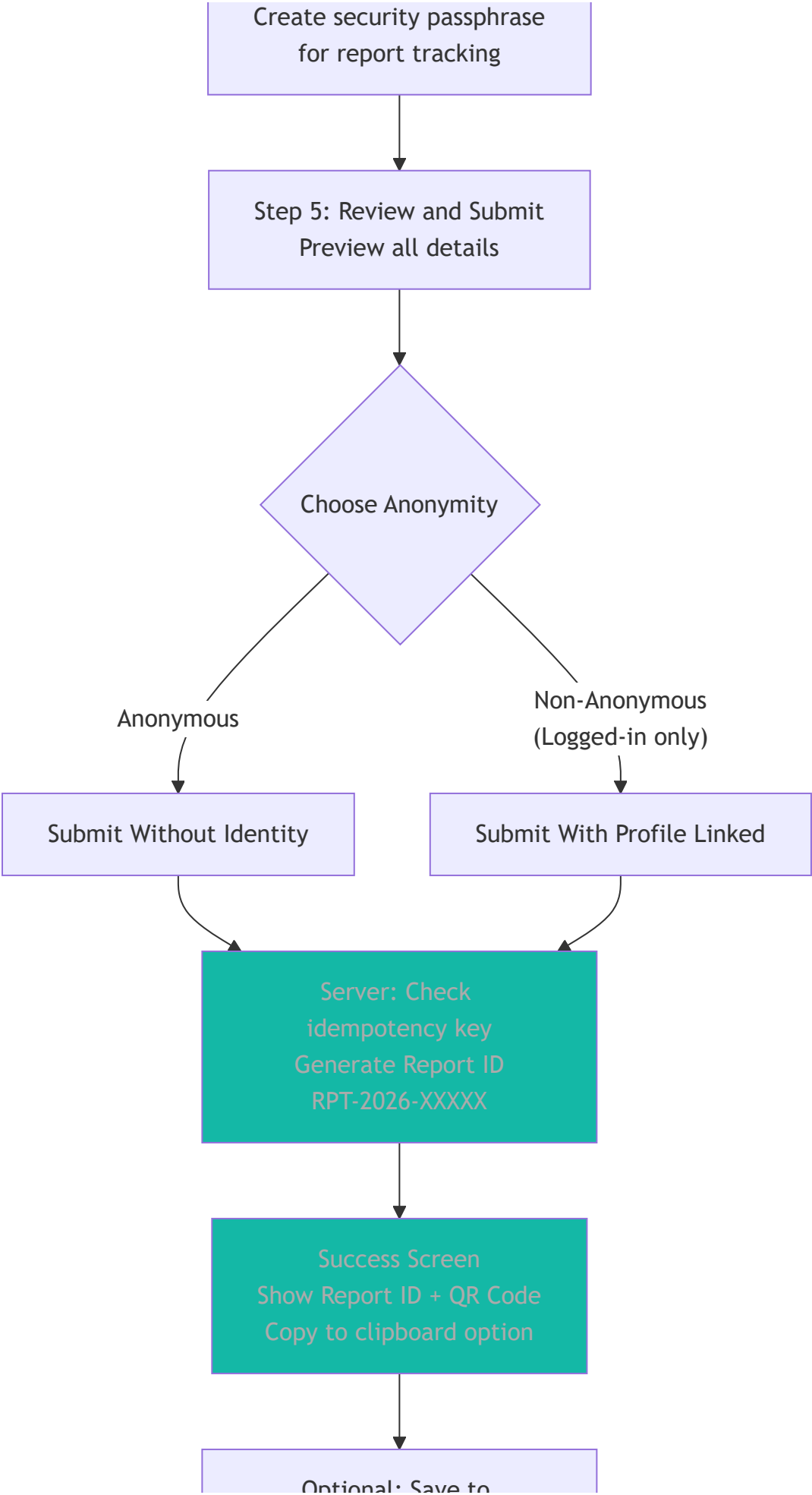
Browser	Minimum Version
Safari	15+
Edge	90+
Samsung Internet	15+
iOS Safari	15+

`backdrop-filter` (glassmorphism) graceful degradation: falls back to solid semi-transparent background on unsupported browsers.

12. User Flows

12.1 Anonymous Report Submission Flow

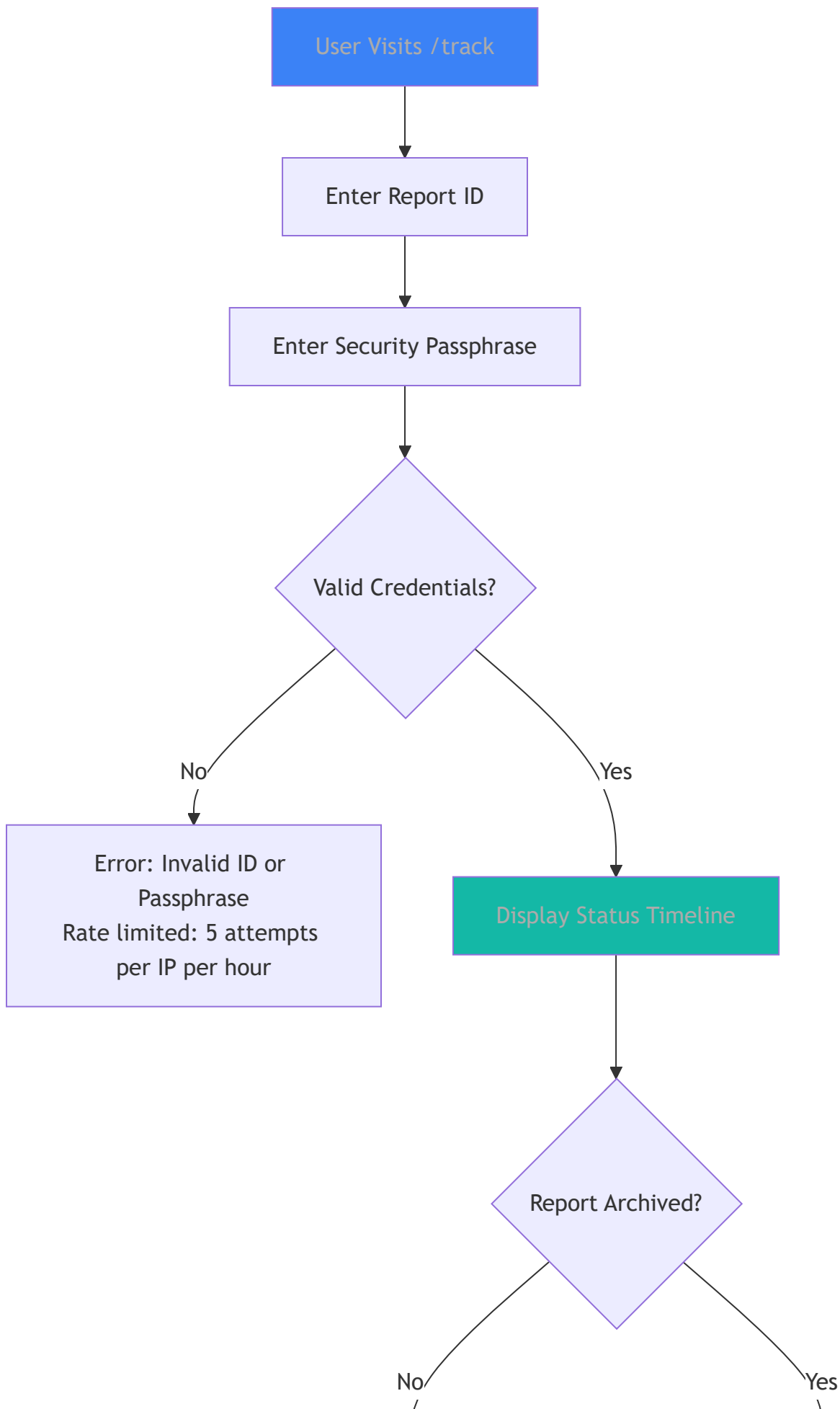


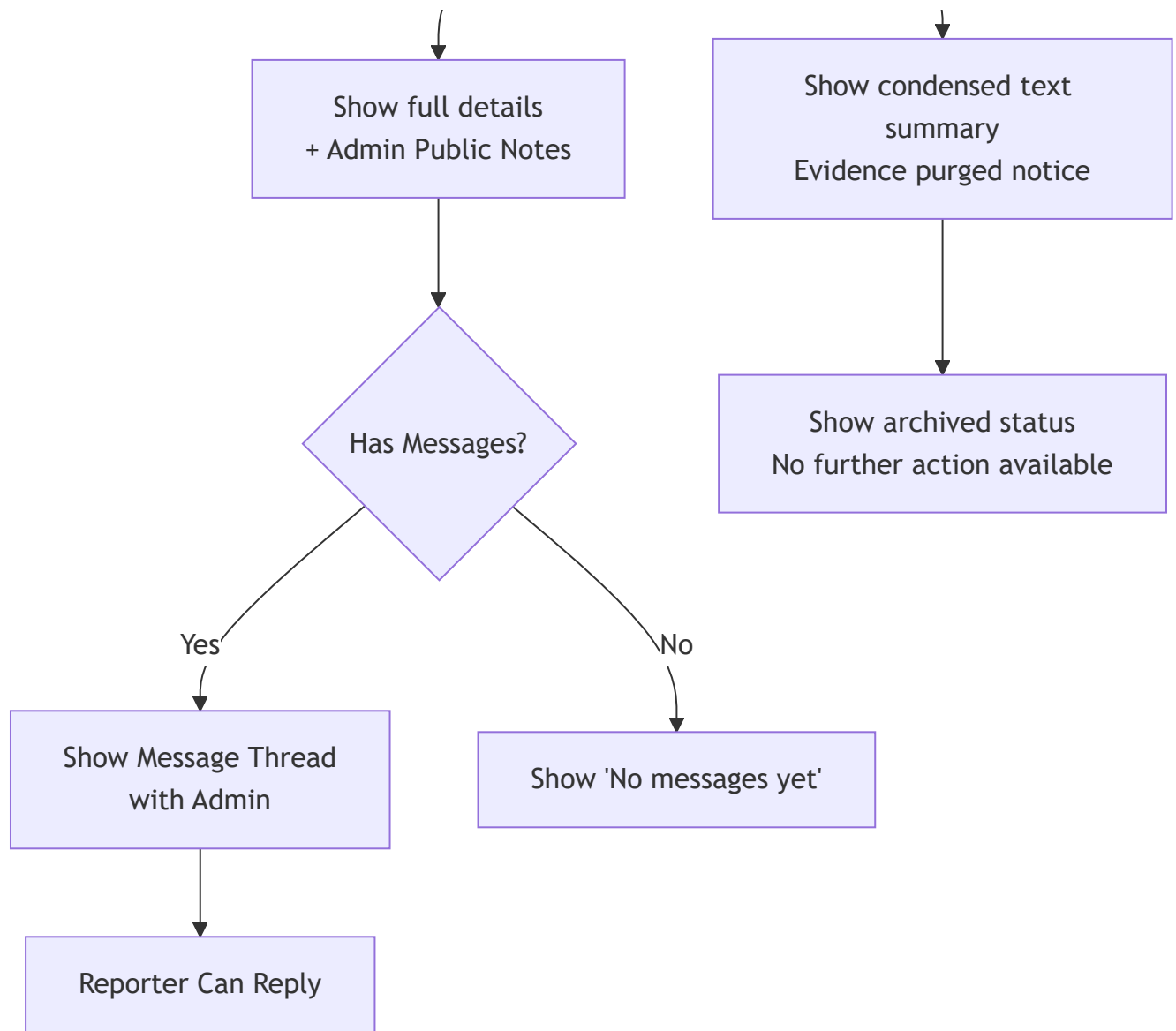


Optional: save to

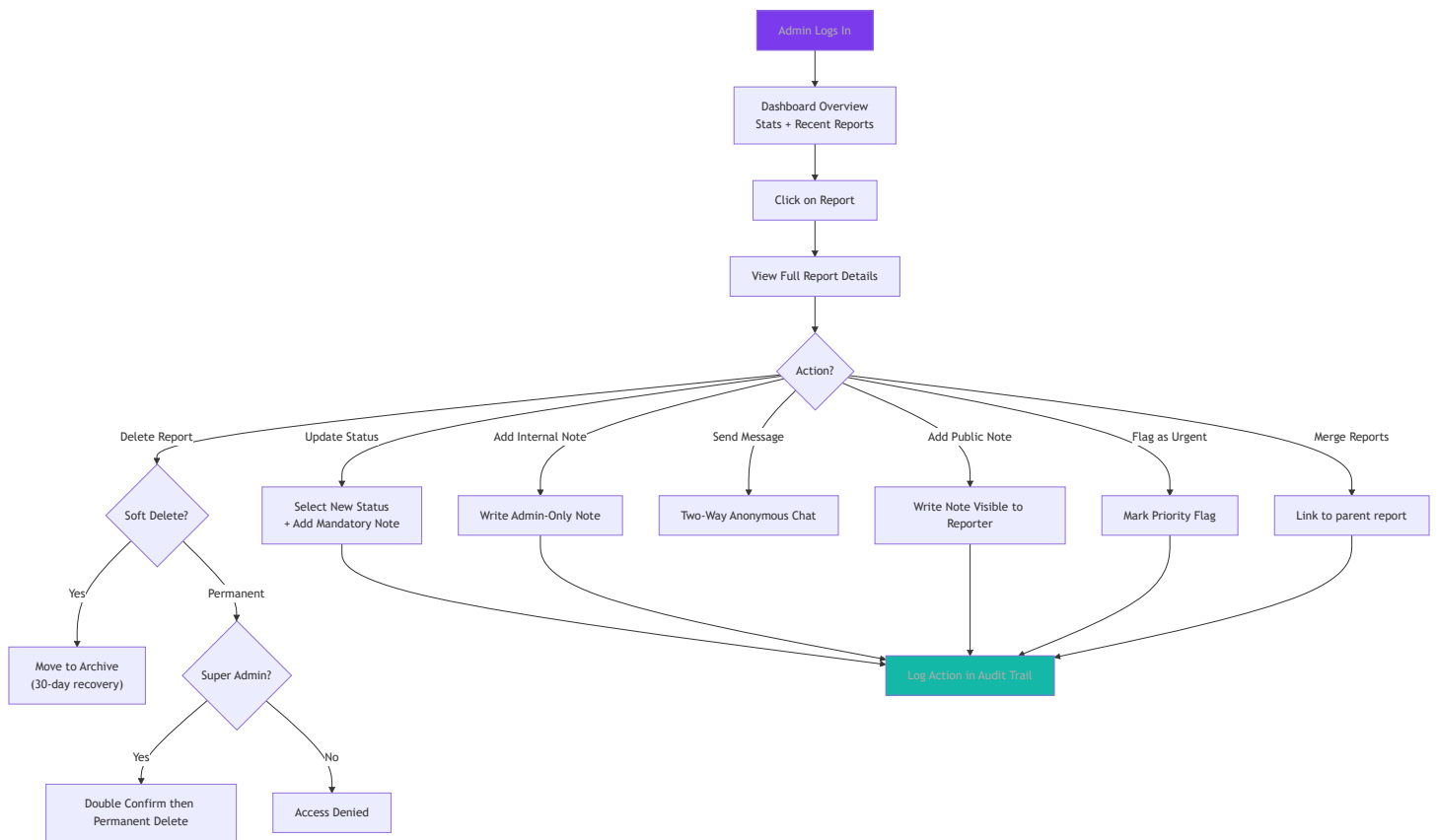
dashboard
(if logged in)

12.2 Report Tracking Flow

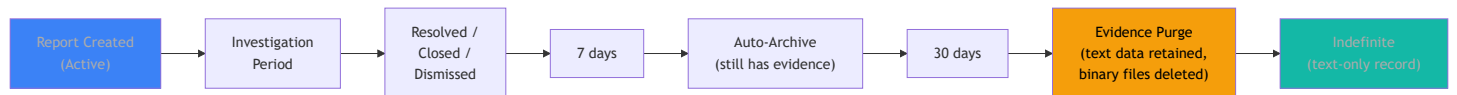




12.3 Admin Report Handling Flow



12.4 Data Lifecycle Flow



13. Edge Cases & Business Logic

13.1 User Edge Cases

Scenario	Handling
User wants to withdraw report	Allowed only if status is Submitted or Received . After Under Review , shows "Request Withdrawal" button which sends a request to admin. Admin can approve or deny. Withdrawal reason is mandatory

Scenario	Handling
User loses Report ID	If logged in, all reports visible on dashboard. If anonymous, no recovery possible by design (anonymity over convenience). Guidance shown prominently on success screen to save the ID
User submits duplicate report	Idempotency key prevents exact double-clicks. Intentional duplicates are allowed; admin can merge or link related reports manually
User submits empty/junk report	Prevented by: mandatory fields, min character limits (50 chars for description), honeypot for bots, rate limiting (5 submissions/hour per IP)
User tries to track with wrong passphrase	5 attempts allowed per IP per hour. After that, 1-hour lockout with friendly message: "Too many attempts. Please try again later."
User uploads unsupported file	Client-side validation: only <code>image/png</code> , <code>image/jpeg</code> , <code>image/webp</code> . Rejected immediately with clear error before upload attempt
User uploads very large image	Client-side compression resizes to max 1920px and converts to WebP. If still > 5MB after compression, rejected with "Image too large even after optimization" message
User on slow connection	Skeleton loaders, optimistic UI updates, auto-retry for failed uploads (3 attempts), progress bar on uploads, graceful timeout message after 30s
User refreshes during multi-step form	Form state saved to localStorage with 7-day TTL, restored on return with "Resume or Start Over?" prompt
Stale localStorage draft	Drafts older than 7 days are auto-cleared on next visit. User sees no stale data
User with JavaScript disabled	Progressive enhancement: basic HTML form still functional via server-side form handling, server-side validation covers all checks
Session expires mid-form	Silent token refresh via Supabase; if refresh fails, form data is preserved in localStorage and user is prompted to re-authenticate
Double-click on submit button	Button disabled immediately on first click + idempotency key prevents server-side duplicate processing

Scenario	Handling
User tries to access archived report via tracking	Shows condensed text-only view with notice: "This case has been resolved. Evidence files have been removed per our data retention policy."
User deletes their account	Anonymous reports are orphaned (no user linked, still accessible via Report ID). Non-anonymous reports have <code>user_id</code> set to null. Profile data permanently deleted
Concurrent submissions from same anonymous session	Each submission gets its own idempotency key; all are accepted independently
User navigates away during upload	Upload cancelled cleanly; partial files cleaned up; draft state preserved for retry
Timezone differences	All timestamps stored in UTC. Client-side renders in user's local timezone via <code>Intl.DateTimeFormat</code> . Admin sees UTC in logs, local time in UI
User tries to message on archived report	Messaging input disabled with notice: "This case is closed. No further messages can be sent."
User submits contact/feedback repeatedly	Rate limited per IP (contact: 3/hr, feedback: 5/day). After limit, shows "You have reached the submission limit. Please try again later."
User sets very short passphrase	Minimum 6 characters enforced client-side and server-side. Error shown: "Passphrase must be at least 6 characters."
User copies Report ID with extra whitespace	Tracking page trims whitespace from input before lookup. No false negatives from accidental spaces

13.2 Admin Edge Cases

Scenario	Handling
Admin tries to delete own admin role	Blocked: at least one Super Admin must exist at all times
Two admins update same report simultaneously	Last-write-wins with conflict notification via real-time subscription; status history preserves both changes with timestamps for audit

Scenario	Handling
Admin tries to access soft-deleted report	Archived reports accessible in "Archive" tab for 30 days. After permanent deletion, returns 404
Admin account compromised	Super Admin can revoke access instantly; all sessions invalidated via Supabase admin API; audit log shows unauthorized actions
Admin removed from whitelist while logged in	Middleware revalidates admin status on every request; removed admin gets immediate 403 and is redirected to a "Access Revoked" page
No admins available	System continues receiving reports; email notifications sent to all admin emails; backlog visible on next login
Bulk action on 100+ reports	Paginated server-side processing with progress indicator; confirmation modal for bulk status changes; CRON handles large batches
Admin tries to change status backwards	Status transitions are enforced: cannot go from <code>Resolved</code> back to <code>Submitted</code> . Only forward transitions or explicit reopen (creates new status entry)
Evidence purge runs during admin viewing	Admin sees real-time update: evidence gallery replaced with "Evidence purged" metadata view. No errors
Analytics with zero data	All chart components handle empty datasets gracefully with illustrated "Not enough data" placeholders instead of broken charts
Admin exports report after evidence purge	PDF export includes text data and metadata. Evidence section shows "Purged on [date]" with original filenames listed
Storage quota approaching limit	Dashboard shows warning banner at 80% usage. At 95%, new uploads are blocked with admin notification. Evidence purge CRON can be triggered manually

13.3 Profanity Filter Decision

Decision: No automatic profanity filter.

Rationale:

- Bullying reports will naturally contain distressing language; filtering it would censor legitimate reports
- False positives would block valid reports and erode user trust
- Admins have full context to assess report content manually
- A content warning banner on the admin panel alerts admins that content may be distressing

14. Email & Notification System

Trigger	Recipient	Channel	Batching
New report submitted	All admins	Email notification	Digest: if 5+ reports in 10 min, consolidate
Report status changed	Reporter (if email provided)	Email notification	Immediate
New message from admin	Reporter (if logged in)	Dashboard badge + email	Immediate
New message from reporter	Assigned admin	Email notification	Immediate
Report flagged as critical	All admins	Priority email	Immediate (bypasses digest)
Weekly analytics digest	Super Admin	Email summary	Weekly (Monday 09:00)
New contact form submission	All admins	Email notification	Digest with report batch
Storage quota warning (80%+)	Super Admin	Email alert	Immediate
Evidence purge completed	Super Admin	Email log	Daily digest
Failed CRON job	Super Admin	Email alert	Immediate

Email bounce handling: Failed sends are retried twice with exponential backoff (5s, 30s). After 2 failures, the failure is logged in `admin_logs` and the admin is notified via dashboard banner on next

login.

15. SEO & Performance

15.1 SEO Strategy

Element	Implementation
Meta Tags	Dynamic <code><title></code> and <code><meta description></code> per page via Next.js metadata API
Open Graph	OG image, title, description for social sharing
Structured Data	JSON-LD for organization schema
Sitemap	Auto-generated <code>sitemap.xml</code> via Next.js
Robots.txt	Allow public pages, disallow admin routes and API
Canonical URLs	Prevent duplicate content issues
Heading Hierarchy	Single <code><h1></code> per page, proper <code><h2></code> to <code><h6></code> nesting
Semantic HTML	<code><main></code> , <code><article></code> , <code><section></code> , <code><nav></code> , <code><footer></code>

15.2 Performance Targets

Metric	Target
First Contentful Paint	< 1.2s
Largest Contentful Paint	< 2.0s
Time to Interactive	< 2.5s
Cumulative Layout Shift	< 0.05
Lighthouse Score	95+ across all categories
Bundle Size	< 150KB initial JS

15.3 Performance Techniques

- Next.js Image Optimization (`<Image>` component with `responsive` `sizes` prop)
- Code splitting via dynamic imports (`next/dynamic` with loading skeletons)
- Server-side rendering for public pages
- Static generation with ISR for resource/about pages (1-hour revalidation)
- Lazy loading for below-the-fold content
- Optimistic UI updates for form submissions
- Supabase connection pooling via Supavisor
- Font subsetting (only Latin characters loaded)
- CSS containment for complex layout components
- `prefers-reduced-motion` media query respected for all animations

16. Project File Structure

```
bullying/
├─ public/
│   ├─ favicon.ico
│   ├─ og-image.png
│   └─ assets/
│       └─ images/
├─ src/
│   ├─ app/
│   │   ├─ layout.js           # Root layout (fonts, metadata, providers)
│   │   ├─ page.js            # Home page
│   │   ├─ globals.css        # Design system (CSS variables, base styles)
│   │   ├─ not-found.js       # Custom 404 page
│   │   ├─ error.js           # Global error boundary
│   │   ├─ loading.js         # Global loading state
│   │   └─ report/
│   │       └─ page.js        # Multi-step report form
│   │   └─ track/
│   │       └─ page.js        # Report tracking
│   │   └─ resources/
│   │       └─ page.js        # Help & resources
│   │   └─ about/
│   │       └─ page.js        # About us
│   │   └─ contact/
│   │       └─ page.js        # Contact form
│   │   └─ feedback/
│   │       └─ page.js        # Feedback form
│   │   └─ login/
│   │       └─ page.js        # Google OAuth login
│   │   └─ auth/
│   │       └─ callback/
│   │           └─ route.js    # OAuth callback handler
│   │   └─ dashboard/
│   │       ├─ layout.js      # User dashboard layout
│   │       ├─ page.js        # User dashboard home
│   │       └─ reports/
│   │           └─ page.js     # User's reports
│   │           └─ profile/
│   │               └─ page.js # Profile management
│   │   └─ admin/
│   │       └─ layout.js      # Admin layout (sidebar)
```

```

├── login/
│   └── page.js # Admin login
├── page.js # Admin dashboard
├── reports/
│   ├── page.js # All reports table
│   └── [id]/
│       └── page.js # Report detail
├── archive/
│   └── page.js # Archived reports
├── analytics/
│   └── page.js # Analytics dashboard
├── users/
│   └── page.js # User management
├── feedback/
│   └── page.js # Feedback viewer
├── contacts/
│   └── page.js # Contact submissions
├── logs/
│   └── page.js # Activity logs
├── settings/
│   └── page.js # Platform settings
├── api/
│   ├── reports/
│   │   └── route.js # Report CRUD API
│   ├── track/
│   │   └── route.js # Report tracking API
│   ├── messages/
│   │   └── route.js # Messaging API
│   ├── feedback/
│   │   └── route.js # Feedback API
│   ├── contact/
│   │   └── route.js # Contact form API
│   ├── cron/
│   │   ├── evidence-purge/
│   │   │   └── route.js # Evidence purge endpoint
│   │   ├── auto-archive/
│   │   │   └── route.js # Auto-archive endpoint
│   │   └── cleanup/
│   │       └── route.js # General cleanup endpoint
│   └── admin/
│       ├── reports/
│       │   └── route.js # Admin report management
│       └── analytics/

```

		└─ route.js	# Analytics data
		└─ users/	
		└─ route.js	# User management
		└─ storage/	
		└─ route.js	# Storage usage stats
		└─ logs/	
		└─ route.js	# Audit logs
	└─ components/		
	└─ ui/		# Reusable UI primitives
	└─ Button.jsx		
	└─ Card.jsx		
	└─ Input.jsx		
	└─ Modal.jsx		
	└─ Badge.jsx		
	└─ Toast.jsx		
	└─ Skeleton.jsx		
	└─ Table.jsx		
	└─ Pagination.jsx		
	└─ Timeline.jsx		
	└─ Accordion.jsx		
	└─ FileUpload.jsx		
	└─ EmptyState.jsx		
	└─ ErrorState.jsx		
	└─ layout/		# Layout components
	└─ Navbar.jsx		
	└─ Footer.jsx		
	└─ Sidebar.jsx		
	└─ MobileNav.jsx		
	└─ report/		# Report form components
	└─ StepIndicator.jsx		
	└─ IncidentTypeStep.jsx		
	└─ DetailsStep.jsx		
	└─ EvidenceStep.jsx		
	└─ SecurityStep.jsx		
	└─ ReviewStep.jsx		
	└─ dashboard/		# User dashboard components
	└─ ReportCard.jsx		
	└─ StatusTimeline.jsx		
	└─ admin/		# Admin-specific components
	└─ StatsCard.jsx		
	└─ StorageMeter.jsx		
	└─ ReportTable.jsx		
	└─ ReportDetail.jsx		

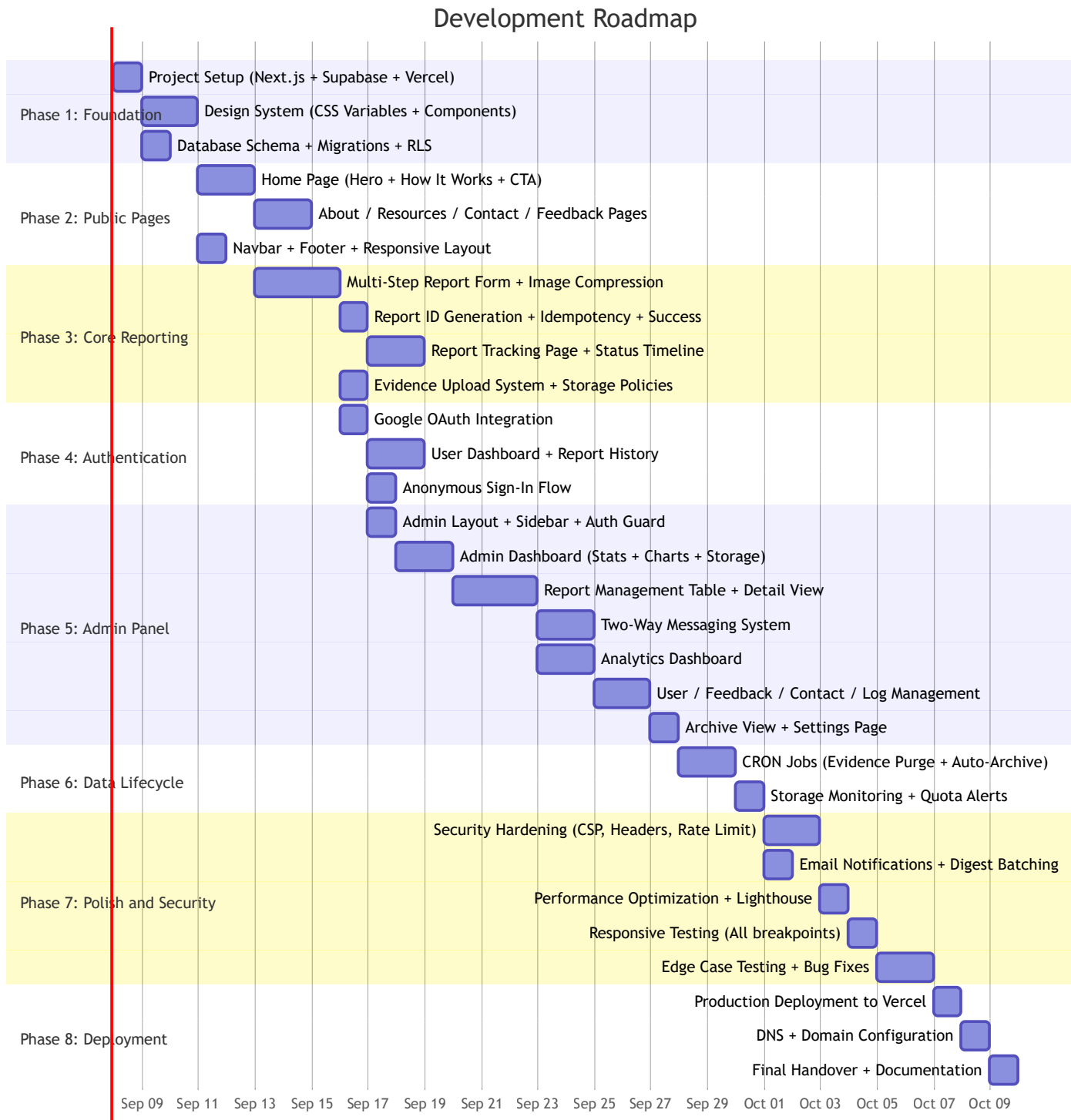
```

| | | └─ MessageThread.jsx
| | | └─ AnalyticsCharts.jsx
| | | └─ UserTable.jsx
| | | └─ ActivityLog.jsx
| | └─ home/                                # Home page sections
| |   └─ Hero.jsx
| |   └─ HowItWorks.jsx
| |   └─ Statistics.jsx
| |   └─ CTA.jsx
| └─ lib/
| | └─ supabase/
| | | └─ client.js                        # Browser Supabase client
| | | └─ server.js                      # Server Supabase client
| | | └─ admin.js                      # Service-role client (server-only)
| | └─ utils/
| | | └─ generateReportId.js           # Cryptographic Report ID generator
| | | └─ hashPassphrase.js            # Passphrase hashing (bcrypt)
| | | └─ compressImage.js             # Client-side image compression
| | | └─ formatDate.js               # Date formatting (UTC to local)
| | | └─ idempotency.js              # Idempotency key generation
| | | └─ validators.js               # Zod schemas
| | └─ constants/
| | | └─ statuses.js                 # Status definitions & valid transitions
| | | └─ categories.js              # Bullying categories
| | | └─ roles.js                   # User role definitions
| | | └─ limits.js                  # Rate limits, file limits, retention periods
| └─ hooks/
| | └─ useAuth.js                    # Authentication hook
| | └─ useReports.js                # Report data fetching with pagination
| | └─ useRealtime.js              # Supabase real-time subscriptions
| | └─ useToast.js                  # Toast notification hook
| | └─ useLocalDraft.js            # localStorage draft management with TTL
| └─ middleware.js                  # Auth guards, rate limiting, CSRF, admin revalidation
└─ supabase/
  └─ migrations/
    └─ 001_initial_schema.sql        # Tables, indexes, RLS policies
    └─ 002_cron_jobs.sql             # pg_cron scheduled jobs
    └─ 003_storage_policies.sql      # Storage bucket policies
  └─ seed.sql                       # Seed data (admin users)
  └─ config.toml                    # Supabase local config
└─ .env.local                       # Environment variables (not committed)
└─ .env.example                     # Template for env vars
└─ next.config.js                   # Next.js config (headers, rewrites, CSP)

```

└─ package.json
└─ README.md

17. Development Phases & Timeline



18. Environment Variables

```
# Supabase
NEXT_PUBLIC_SUPABASE_URL=
NEXT_PUBLIC_SUPABASE_ANON_KEY=
SUPABASE_SERVICE_ROLE_KEY=          # Server-only, NEVER exposed to client

# Google OAuth (configured in Supabase dashboard)
# No separate env vars needed, handled by Supabase Auth

# Email (Resend)
RESEND_API_KEY=

# App
NEXT_PUBLIC_SITE_URL=
NEXT_PUBLIC_SITE_NAME=TBA

# Admin whitelist (comma-separated emails)
ADMIN_EMAILS=
SUPER_ADMIN_EMAILS=

# CRON Security (shared secret to authenticate cron endpoints)
CRON_SECRET=

# Rate Limiting
RATE_LIMIT_WINDOW_MS=60000
RATE_LIMIT_MAX_REQUESTS=30
```

19. Verification & Quality Assurance

19.1 Automated Testing

Type	Tool	Coverage
Unit Tests	Vitest	Utility functions, validators, ID generation, compression

Type	Tool	Coverage
Component Tests	React Testing Library	Form components, UI interactions, empty states
Integration Tests	Vitest + Supabase test helpers	API routes, RLS policies, CRON jobs
E2E Tests	Playwright	Full user flows (report > track > admin > archive > purge)

19.2 Manual Testing Checklist

- ☐ Anonymous report submission flow (end-to-end)
- ☐ Authenticated report submission flow
- ☐ Report tracking with correct ID + passphrase
- ☐ Report tracking with wrong credentials (rate limiting kicks in after 5 attempts)
- ☐ Report tracking for archived/purged report (shows text-only view)
- ☐ Admin login with whitelisted email
- ☐ Admin login with non-whitelisted email (rejected)
- ☐ Admin removed mid-session (immediate access revocation)
- ☐ Admin status update reflected on reporter tracking page
- ☐ Admin two-way messaging
- ☐ Admin soft-delete and recovery within 30 days
- ☐ Admin permanent delete (Super Admin only, double confirmation)
- ☐ Admin merge duplicate reports
- ☐ Admin bulk actions (status change, archive, export)
- ☐ File upload: valid image types accepted and compressed
- ☐ File upload: oversized images compressed successfully
- ☐ File upload: invalid types rejected client-side
- ☐ Evidence purge CRON: evidence deleted 30 days after resolution
- ☐ Auto-archive CRON: resolved reports archived after 7 days
- ☐ Storage meter accuracy on admin dashboard
- ☐ Email notifications: new report, status change, critical flag
- ☐ Email digest batching: 5+ reports in 10 min = 1 email
- ☐ Email bounce handling: failures logged, not retried indefinitely
- ☐ Double-click submit prevention (idempotency)
- ☐ localStorage draft resume after browser close
- ☐ localStorage draft expiry after 7 days

- ☐ Timezone display: UTC stored, local time shown to user
- ☐ Mobile responsiveness: all pages at 320px, 480px, 768px, 1024px, 1440px
- ☐ Tablet responsiveness: landscape and portrait orientations
- ☐ Keyboard navigation: all interactive elements accessible via Tab
- ☐ Screen reader: all form fields properly labelled
- ☐ prefers-reduced-motion : animations disabled when preference set
- ☐ Glassmorphism fallback: solid background on unsupported browsers
- ☐ Empty state handling: dashboard, analytics, reports, feedback all show proper empty states
- ☐ Error boundary: broken components don't crash the entire app
- ☐ 404 page: custom styled, not default Next.js
- ☐ Lighthouse audit: 95+ scores across all categories
- ☐ Security headers: verified via securityheaders.com
- ☐ Google OAuth down: graceful error message shown

20. Deliverables Summary

#	Deliverable	Description
1	Production Website	Fully functional, deployed on Vercel with custom domain
2	Admin Dashboard	Custom admin panel with full case management and analytics
3	Database	Supabase PostgreSQL with RLS, migrations, seed data, and CRON jobs
4	Data Lifecycle System	Automated evidence purge, auto-archive, and storage monitoring
5	Source Code	Clean, documented, production-ready codebase
6	Documentation	Setup guide, admin manual, environment configuration
7	Analytics	Built-in reporting analytics dashboard with export
8	Email System	Automated notifications with digest batching and bounce handling
9	Security	Enterprise-grade security hardening across all 5 layers
10	QA Report	Test results from all automated and manual test cases

This document serves as the single source of truth for the entire project. Every feature, design decision, security measure, storage optimization, and technical choice is documented above. Development begins upon approval.